

HRIDAYA SHRESTHA

Pancreatic Cancer Researcher

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Summary

- PhD graduate with over six years of experience in pancreatic cancer research using genetically engineered mouse models.
- Skilled in molecular and cell biology techniques for analyzing gene and protein expression, with emphasis on cancer cell signaling pathways.
- Proven track record of developing and executing research projects, publishing research papers, and presenting at conferences.

Research Interests

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|------------------------------|--------------------------|-----------------------------|
| • Cancer signalling pathways | • Pancreatic cancer | • Mouse models of cancer |
| • Pancreatitis | • Tumor microenvironment | • Oncology drug development |

Research Experience

Postdoctoral Researcher

Stony Brook University | Stony Brook, NY

03/2026 - present

Mentor: Prof. Mario Shields

Project: *Role of Gal3 signaling in modulating the tumor immune microenvironment and drug resistance*

Postdoctoral Researcher

Wayne State University | Detroit, MI

02/2018 - 06/2024

Mentor: Prof. Kay-Uwe Wagner

Project: *Inflammatory cytokine signaling in pancreas development, pancreatitis and pancreatic cancer*

- Developed and analyzed genetically engineered mouse models of pancreatic cancer, including long-term phenotypic studies, derivation of primary tumor cell lines, and downstream molecular analyses (RT-qPCR, RNA-Seq, Western blot, ELISA).
- Designed and executed CRISPR knockouts, siRNA/shRNA knockdowns, and retroviral/lentiviral-based gene overexpression experiments in pancreatic cancer cell lines.
- Performed *in vitro* cellular assays (viability, proliferation, migration, spheroid formation) and *in vivo* studies (CDX/PDX tumor implantation, growth assessment) to analyze JAK1/STAT3 signaling in cancer progression.
- Performed histological analyses, including immunohistochemistry and immunofluorescence (IHC/IF) staining on mouse pancreatic tissues and quantitative imaging for comparative pathology.
- Established the setup for pancreatic acinar cell culture to assess JAK1's role in acinar-to-ductal metaplasia (ADM).
- Performed bulk RNA-seq studies, including RNA extraction and bioinformatic analysis, to determine the role of JAK1 in pancreas development, acute and chronic pancreatitis, and cancer progression.
- Mentored trainees and contributed to publications and lab operations, including first-author and co-authored papers, collaborative projects on mammary gland development and tumorigenesis, and laboratory management.

Graduate Student

Chonnam National University | Gwangju, South Korea

08/2012 - 08/2017

Project: *Factors affecting adherens junction proteins in prostate cancer*

- Performed cell culture and molecular techniques, including molecular cloning, Western blot, immunoprecipitation, cell surface protein biotinylation, RT-PCR, and cell-based functional assays.
- Studied trafficking of adherens junction proteins in cancer cells using live and fixed cell confocal microscopy.
- Led three independent research projects on adherens junction regulation, resulting in three first-author publications, two co-authored papers, and multiple conference presentations.

Additional Experience

Lecturer

Pinnacle Technical College Pvt. Ltd. | Lalitpur, Nepal

11/2008 - 01/2010

- Mentored and supervised Diploma in Pharmacy (CTEVT Board of Nepal) students through conducting theory classes and overseeing practical lessons in Pharmacology.

Regulatory Affairs Pharmacist

Department of Drug Administration, Ministry of Health and Population, Government of Nepal

01/2010 - 07/2010

- Conducted sample collection, inspections, and inquiries at pharmacy outlets across various regions of the country to ensure compliance with regulatory standards.

Education

Doctor of Philosophy in Pharmacy (Molecular Pathology)

Chonnam National University | Gwangju, South Korea

09/2014 - 08/2017

- **Advisor:** Prof. Kwonseop Kim
- **GPA:** 4.46/4.5
- **Dissertation title:** Investigation of the molecular mechanism of Hakai-mediated stabilization of δ -catenin

Master of Pharmacy

Chonnam National University | Gwangju, South Korea

08/2012 - 08/2014

- **Advisor:** Prof. Kwonseop Kim
- **GPA:** 4.29/4.5
- **Thesis title:** Hakai, an E3-ligase for E-cadherin, stabilizes δ -catenin and affects E-cadherin complex

Bachelor of Pharmaceutical Sciences

Pokhara University | Kaski, Nepal

08/2004 - 08/2008

- **GPA:** 3.59/4.0
- **Thesis title:** Study on Traditional Medicines of Western Nepal

Awards, Scholarships and Fellowships

WooJung Education and Culture Foundation Aid Fund

WooJung Education and Culture Foundation, South Korea

2015 - 2017

Fostering Next Generation Researchers Program Scholarship

Chonnam National University, Gwangju, South Korea

2013 - 2017

Korean Government Academic Achievement Award

National Institute of Education and Development, Ministry of Education, Science and Technology, South Korea

2013

Korean Government Scholarship

National Institute of Education and Development, Ministry of Education, Science and Technology, South Korea

2010 - 2014

Training and Workshops

Research Internship in Cancer Genomics Genomac Institute Inc. Remote	01/2025 – 04/2025
Workshop on Genomes, Networks & Pathways (5th Ed.) Decode Life Remote	09/2024 - 10/2024
Data Management & Statistical Analysis in R Workshop Wayne State University Detroit, MI	01/2021 - 05/2021

Skills

- ***In vivo* animal study (mouse):** Handling and restraining animals, tipping tails, tagging ears, administering drugs (intravenous, subcutaneous, and intraperitoneal injections), monitoring health, euthanasia, anesthesia, *in vivo* bioluminescence imaging, necropsy, tissue collection (lymph nodes, spleen, thymus, pancreas, mammary glands, lungs, liver, prostate), cell isolation, digestion of mammary glands for flow cytometry, surgical transplantation of cancer cells.
- **Tissue culture:** Mammalian cell culture, cell line development, clonal isolation and expansion, stable and transient transfections, retroviral and lentiviral-based gene overexpression, electroporation, 3D pancreatic acinar cell culture, RNAi knockdown, CRISPR knockout, cell-based functional assays such as cytotoxicity and cell viability assessments.
- **Molecular biology:** Molecular cloning, DNA/RNA/protein isolation, RT-qPCR, Western blot, ELISA
- **Microscopy:** Brightfield and fluorescence microscopy, immunocytochemistry (ICC), immunohistochemistry (IHC), immunofluorescence (IF)
- **Software and Data Analysis:**
 - **Bioinformatics:** Bulk and single-cell RNA-Seq analysis
 - **Software:** ImageJ, QuPath, GraphPad Prism, R, SnapGene, EndNote, Microsoft Office Suite, Adobe Illustrator/Photoshop
- **Soft skills:** Communication, attention to detail, teamwork, leadership

Professional Societies

Associate Member American Association for Cancer Research (AACR)	2017- present
Member American Pancreatic Association	2026 - present

Service for Peer-Reviewed Journals

Review of Manuscripts

<i>Scientific Reports</i>	2024 - present
<i>Cellular Signalling</i>	2021 - present
<i>Cancer Control</i>	2020 - present
<i>The International Journal of Biochemistry & Cell Biology</i>	2021 - present
<i>European Journal of Cell Biology</i>	2022 - present
<i>Cells</i>	2022 - present
<i>Cancers</i>	2021 - present
<i>International Journal of Molecular Sciences</i>	2021 - present
<i>Biomolecules</i>	2021 - present

Selected Publications

Peer-reviewed journal articles

Reports of original work

1. **Shrestha, H.**, Rädler, P. D., Dennaoui, R., Wicker, M. N., Rajbhandari, N., Sun, Y., Peck, A. R., Vistisen, K., Triplett, A. A., Beydoun, R., Sterneck, E., Saur, D., Rui, H., & Wagner, K. U. (2024). The Janus kinase 1 is critical for pancreatic cancer initiation and progression. *Cell reports*, 43(5), 114202. <https://doi.org/10.1016/j.celrep.2024.114202>
2. Rädler, P. D., Wehde, B. L., Triplett, A. A., **Shrestha, H.**, Shepherd, J. H., Pfefferle, A. D., Rui, H., Cardiff, R. D., Perou, C. M., & Wagner, K. U. (2021). Highly metastatic claudin-low mammary cancers can originate from luminal epithelial cells. *Nature communications*, 12(1), 3742. <https://doi.org/10.1038/s41467-021-23957-5>
3. Sakamoto, K., Rädler, P. D., Wehde, B. L., Triplett, A. A., **Shrestha, H.**, Ferraiuolo, R. M., Amari, F., Coppola, V., Klinakis, A., Efstratiadis, A., & Wagner, K. U. (2020). Efficient tissue-type specific expression of target genes in a tetracycline-controlled manner from the ubiquitously active Eef1a1 locus. *Scientific reports*, 10(1), 207. <https://doi.org/10.1038/s41598-019-57052-z>
4. Wehde, B. L., Rädler, P. D., **Shrestha, H.**, Johnson, S. J., Triplett, A. A., & Wagner, K. U. (2018). Janus Kinase 1 Plays a Critical Role in Mammary Cancer Progression. *Cell reports*, 25(8), 2192–2207.e5. <https://doi.org/10.1016/j.celrep.2018.10.063>
5. Shrestha, N.*, **Shrestha, H.***, Ryu, T., Kim, H., Simkhada, S., Cho, Y. C., Park, S. Y., Cho, S., Lee, K. Y., Lee, J. H., & Kim, K. (2018). δ -Catenin Increases the Stability of EGFR by Decreasing c-Cbl Interaction and Enhances EGFR/Erk1/2 Signaling in Prostate Cancer. *Molecules and cells*, 41(4), 320–330. [*First authorship shared] <https://doi.org/10.14348/molcells.2018.2292>
6. **Shrestha, H.**, Ryu, T., Seo, Y. W., Park, S. Y., He, Y., Dai, W., Park, E., Simkhada, S., Kim, H., Lee, K., & Kim, K. (2017). Hakai, an E3-ligase for E-cadherin, stabilizes δ -catenin through Src kinase. *Cellular signalling*, 31, 135–145. <https://doi.org/10.1016/j.cellsig.2017.01.009>
7. **Shrestha, H.**, Yuan, T., He, Y., Moon, P. G., Shrestha, N., Ryu, T., Park, S. Y., Cho, Y. C., Lee, C. H., Baek, M. C., Cho, S., Simkhada, S., Kim, H., & Kim, K. (2016). Investigation of the molecular mechanism of δ -catenin ubiquitination: Implication of β -TrCP-1 as a potential E3 ligase. *Biochimica et biophysica acta*, 1863(9), 2311–2321. <https://doi.org/10.1016/j.bbamcr.2016.06.006>
8. He, Y., Ryu, T., Shrestha, N., Yuan, T., Kim, H., **Shrestha, H.**, Cho, Y. C., Seo, Y. W., Song, W. K., & Kim, K. (2016). Interaction of EGFR to δ -catenin leads to δ -catenin phosphorylation and enhances EGFR signaling. *Scientific reports*, 6, 21207. <https://doi.org/10.1038/srep21207>

Review articles

1. Dennaoui, R., **Shrestha, H.**, & Wagner, K. U. (2021). Models of pancreatic ductal adenocarcinoma. *Cancer metastasis reviews*, 40(3), 803–818. <https://doi.org/10.1007/s10555-021-09989-9>

For a complete list of publications, see my **Google Scholar** profile at <https://scholar.google.com/citations?user=eurUHUYAAAAJ>.

Conferences and Presentations

1. **Shrestha, H.**, Ryu, T., Seo, Y. W., Park, S. Y., He, Y., Dai, W., Park, E., Simkhada, S., Kim, H., Lee, K., and Kim, K. “Hakai, an E3-ligase for E-cadherin, stabilizes δ -catenin through Src kinase.” Annual Meeting of the American Association for Cancer Research; Washington, DC., 2017.
2. **Shrestha, H.**, He, Y., Lee, K., and Kim, K. “Hakai, an E3-ligase for E-cadherin, stabilizes δ -catenin and affects E-cadherin.” International Conference of the Korean Society for Molecular and Cellular Biology, Seoul, Korea, 2013.

References

Available upon request.